NHS Severe Patient Harm Forecasting

SPHERE-PPL Contest Submission Report

Daniela Díaz Catalán Ph.D. and Patricio Alarcón Morales

3 July 2026

Introduction

This report presents the deployment-ready machine learning framework developed for the SPHERE-PPL Forecasting Contest. The objective is to predict daily estimated avoidable deaths across the Bristol NHS healthcare system resulting from Emergency Department (ED) admission delays.

The pipeline was designed to respect all operational and data constraints: predictions utilize information recorded up to the strict midday cutoff on day D, and all past target-derived records are strictly bound to a minimum 3-day reporting lag (lags ≥ 3 days). This architectural choice guarantees complete isolation from look-ahead bias and simulates real-world implementation where recent ground truth values are structurally unavailable.

Reproducibility note: The report reads only precomputed outputs from `submission/development/` and `submission/validation/`. Run `Rscript run_forecast_xgb.R development` and `Rscript run_forecast_xgb.R validation` before knitting.

Development Results

Data Preparation

The historical development stream spanning from March 2023 to September 2025 was structuralized into consistent daily matrices. To enforce clinical and operational fidelity, raw metrics recorded prior to 12:00 PM were compiled within the current calendar day, while any updates occurring after midday were dynamically mapped onto the subsequent forecasting day. Missing operational attributes were resolved using a rule-based sequence combining last-observation-carried-forward (LOCF) and column-wise median fallbacks, completely preventing data leakage across rolling origins.

Table 1: Prepared modelling dataset: development

n_days	n_predictors	start_date	end_date
916	360	2023-03-30	2025-09-30

Feature Engineering

A high-dimensional feature space was generated to capture complex non-linear operational dynamics and seasonal fluctuations. The candidate pool includes calendar variables, rolling summaries, safe target historical lags, and multi-order interaction terms designed to isolate severe hospital saturation thresholds.

To achieve optimal generalization under significant temporal drift, structural regularisation was enforced natively within a gradient boosted decision tree framework (XGBoost). Furthermore, a robust safety filter was integrated into the workflow to clip numerical instability or mathematical overflows caused by complex interactions before data writing.

Table 2: Number of predictors by transformation type: development

transformation	n_predictors
None (Raw Operational Indicators)	180
Rolling Windows Summaries	172
Target Lags (>= 3 days)	4
High-Order Interactions	4

Backtesting and Retrospective Validation

Model validation was performed utilizing a strict rolling-origin framework with an expanding training window and a fixed 10-day forward horizon. This approach mimics daily implementation across successive operational windows.

Table 3: Overall Backtesting Mean Squared Error (MSE) by Forecast Horizon

model	mse_1_5	mse_6_10
Hybrid XGBoost (development)	0.0158	0.0175

Table 4: Computational Efficiency and Feature Tracking: Development Windows

Training Window	Horizon	Windows	Total Runtime (min)	Mean Runtime (sec)	Median Runtime (sec)	Mean Features
Dynamic Hybrid (Tuned History)	10 days	151	14.17	5.63	5.61	360

The model demonstrates excellent stability across all retrospective testing periods. Notably, the prediction error shows a remarkably controlled degradation between the short-term horizon (days 1–5) and the long-term horizon (days 6–10), with the average MSE increasing by only 10.7%.

Table 5: Window-level MSE distribution: development

Horizon	Mean MSE	Median MSE
1-5 days (Short-term)	0.0158	0.0135
6-10 days (Long-term)	0.0175	0.0159

Validation Results

The exact preprocessing and modeling script was applied to the validation dataset covering the extended evaluation period. To protect reproducibility, validation outputs were written to a separate physical path, ensuring that hyperparameters and structural components remained unaltered.

Because the assessment dataset holds out the true values of estimated avoidable deaths, error metrics (MSE) cannot be extracted for this phase.

Validation Data Summary

Table 6: Prepared modelling dataset: validation

n_days	n_predictors	start_date	end_date
1097	352	2023-03-30	2026-03-30

Validation Feature Space & Execution Performance

The dimensions of the feature transformations scale consistently with the properties of the raw operational inputs, reflecting a clean pipeline execution.

Table 7: Feature Space Breakdown by Transformation Type: Validation

Transformation Type	Number of Predictors
None (Raw Operational Indicators)	176
Rolling Windows Summaries	168
Target Lags (>= 3 days)	4
High-Order Interactions	4

Table 8: Computational Efficiency and Feature Tracking: Validation Windows

Training Window	Horizon	Windows	Total Runtime (min)	Mean Runtime (sec)	Median Runtime (sec)	Mean Features
Dynamic Hybrid (Tuned History)	10 days	131	10.76	4.93	4.93	352

Comparative Framework and Structural Analysis

The comparative summary outlines structural variations across execution environments. The development pipeline processes a 916-day historical baseline generating 151 rolling origins, whereas the validation phase handles an expanded 1097-day timeline yielding 131 sliding forecast windows. This variance reflects the larger initial historical span required to establish the training window before launching the first out-of-sample ex-ante test point in the validation phase.

Table 9: Performance, Dimensionality and Efficiency Comparison Matrix

Dataset	MSE 1-5d	MSE 6-10d	Total Forecasts	Total Runtime (min)
development	0.0158	0.0175	151	14.17
validation	-	-	131	10.76

An analytical evaluation of the final validation trajectory matrix reveals a distinct bifocal correlation architecture. Forecasts for days 1 to 5 maintain extreme inner correlation ($\rho \approx 0.80 - 0.94$), relying heavily on immediate autoregressive operational drivers. Conversely, days 6 to 10 form a tight independent block ($\rho \approx 0.96 - 0.99$).

A structural drop in correlation occurs precisely at the day 5 to day 6 boundary ($\rho \approx 0.25$). This behavior reflects the mathematical enforcement of the 3-day target reporting lag restriction. Beyond day 5, immediate autoregressive indicators are completely exhausted, forcing the model to seamlessly shift its regime toward high-order structural interactions and macro-capacity indicators. This dual-regime transition demonstrates that the model suppresses random noise and relies on robust, long-term trends for advanced horizons.

Submission Output

The workflow successfully generates a validated 10-day ahead forecast matrix (`pred_matrix.csv`) satisfying all submission specifications. The low average runtime per window (~5 seconds) guarantees full compliance with the competition's 1-hour time constraint, ensuring seamless integration into daily Bristol NHS management routines to support proactive clinical mitigation strategies.